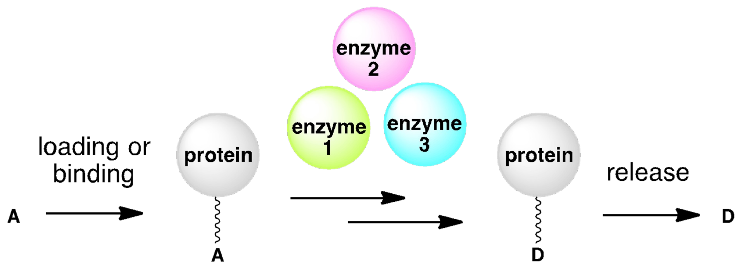
Chemistry 27

Lecture 13: Assembly Line Enzymology:
Fatty Acid Biosynthesis I



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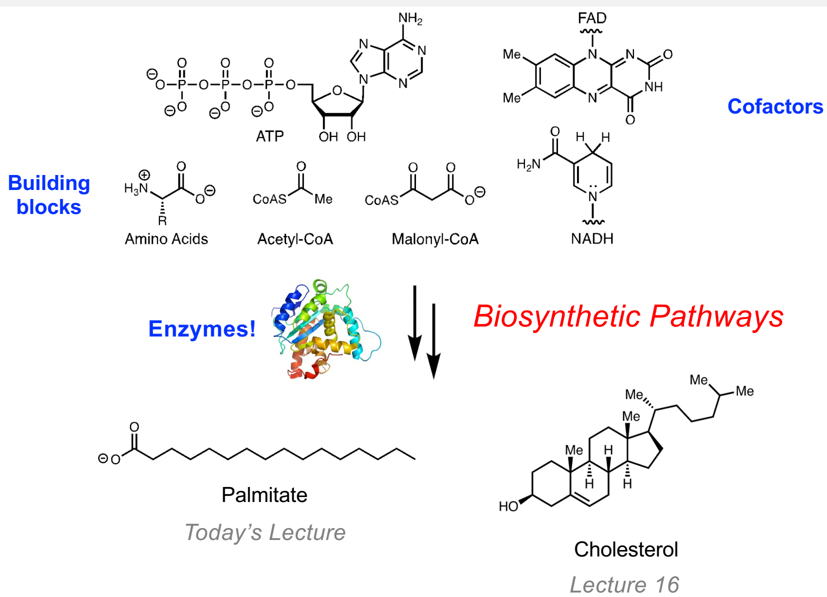
Sirinya Matchacheep, Brian Liau

Introduction to Pathways – Biosynthesis



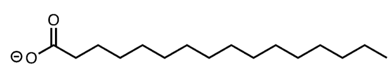
Lecture 13 Overview

- **Fatty acid biosynthesis and assembly line logic**
  - The building blocks
  - Review: Claisen Condensations
  - The assembly line machine and its components
- **Fatty acid biosynthesis arrow pushing**
  - The elongation phase

Introduction to Fatty Acids

- **Fatty acids** are carboxylic acids with hydrocarbon chains.

  Saturated: contains no double bonds
  Unsaturated: contains double bonds



They can exist as straight chains or branched.

Most commonly, fatty acids have an even number of carbons between 12-24.

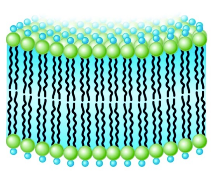
Fatty acid derivatives serve a multitude of functions in the body:

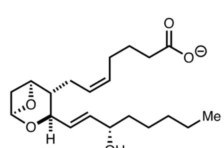
Lipid bilayers

Stored fuels

Signaling molecules

Pigments

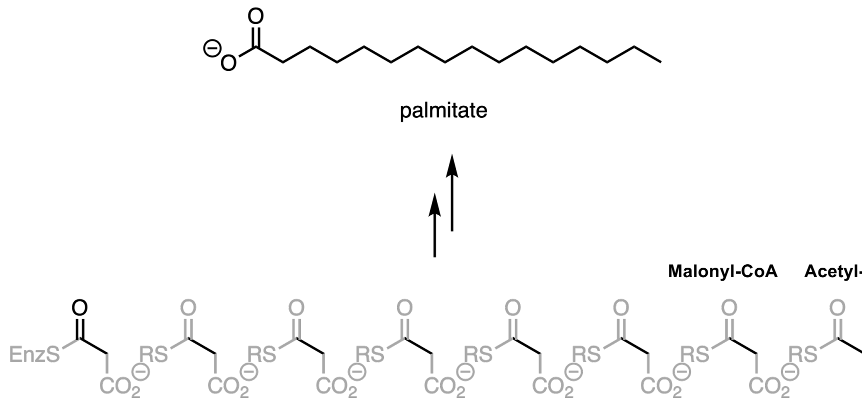




Thromboxane A_2

Fatty Acids are Biosynthesized from Building Blocks of Acetyl-CoA and Malonyl-CoA

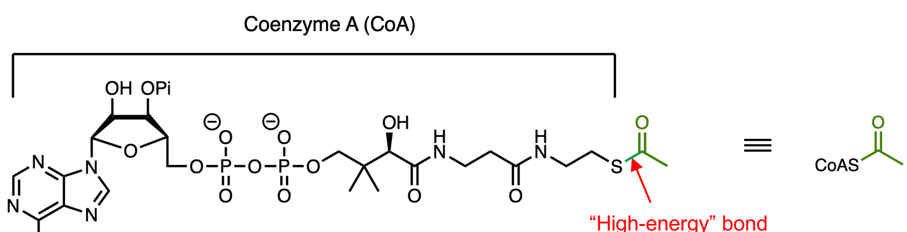
- **Fatty acids** are constructed from simpler building blocks. Notice that the building blocks are at higher oxidation states; **reduction chemistry will be needed**.



Nature uses ***Claisen condensations*** *to put these building blocks together and does so using* ***Enzymatic Assembly Lines****!*

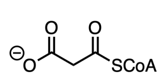
The Parts: Fatty Acid Synthesis Monomers

- Assembly line enzymes always begin with a starter unit. In Fatty Acid Synthase (FAS), the enzyme that makes fatty acids, the starter unit is **acetyl-CoA**.



Acetyl-CoA

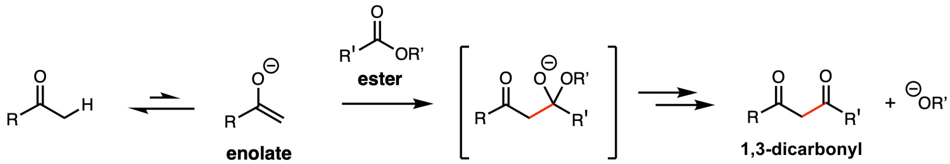
- The monomers added to propagate chain growth are **malonyl-CoA**.



Malonyl-CoA

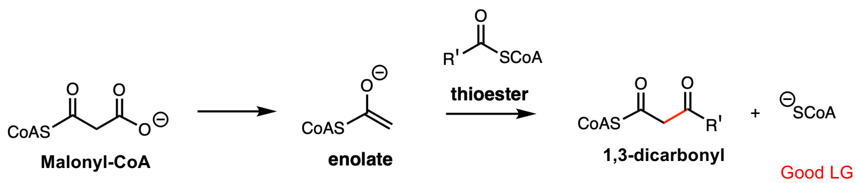
The Key Reaction: Claisen Condensation

Claisen Condensation: enolate (nucleophile) + ester (electrophile)



Challenges under physiological conditions:

Decarboxylative Claisen Condensation: malonyl-CoA (nucleophile) + thioester (electrophile)



Good LG

CO_2 formation drives enolization!

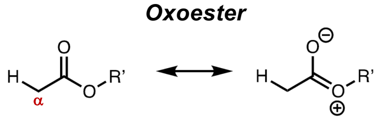
Abbreviated mechanism for Claisen and acyl transfer reactions

Decarboxylative Claisen condensation *allows for facile enolate formation and C-C bond formation, as well as provides an electrophilic functional group for further modification.*

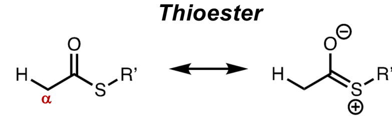
Nature Uses Thioesters to Perform Claisen Condensations

- Due to orbital size mismatch, resonance donation in a thioester is *weaker* than that in an oxoester.

Oxoester



Thioester



Resonance structure makes less of a contribution

Consequences:

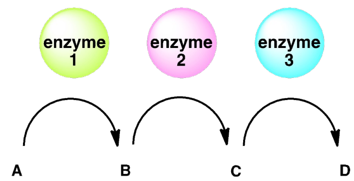
1. Thioesters are more electrophilic than oxoesters and contain high-energy bonds.
2. The α-proton is more acidic on

Poll Everywhere: Compare the acidity of the two alpha protons. Which is more acidic?

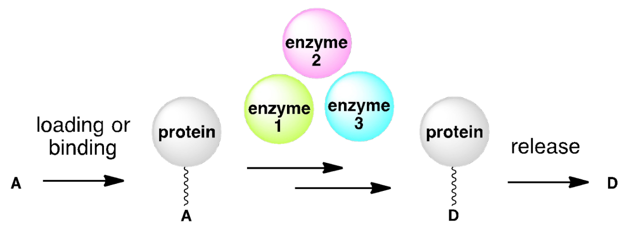
Assembly Line Machinery

- Nature uses enzymatic assembly lines to synthesize fatty acids.

Most metabolic pathways:
Metabolites freely diffuse between enzyme active sites



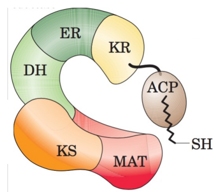
Enzymatic assembly lines:
Multiple chemical modifications occur on a sequestered substrate



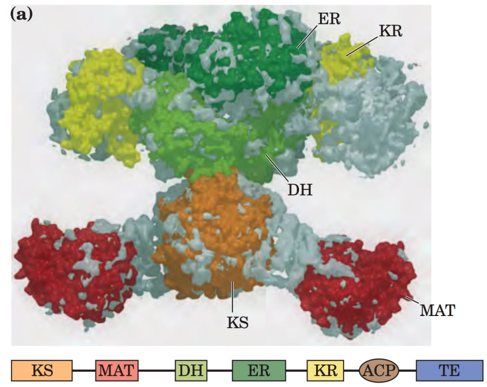
Assembly Line Machinery

Mammalian Fatty Acid Synthase

Fatty Acid Synthases (FAS)



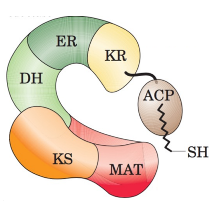
All reactions take place on the single multi-domain enzyme.



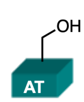
- Very large, multifunctional assembly-line enzymes composed of multiple catalytic domains
- Functions as a homodimer, with each monomer ~272 kDa in mammals
- Catalyzes de novo fatty acid synthesis, producing primarily palmitate (16 carbons)
- Each domain carries out a distinct enzymatic reaction in the fatty acid synthesis cycle

Fatty Acid Synthase Domains

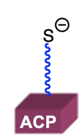
Fatty Acid Synthases (FAS)



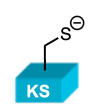
Elongation Domains elongate the fatty acid by 1 unit (2 carbons) via a **decarboxylative Claisen condensation**.



Acyl Transferase

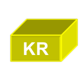


Acyl Carrier Protein



Keto Synthase

Tailoring Domains reduce the ketone of the previous unit to a methylene group in the fatty acid.

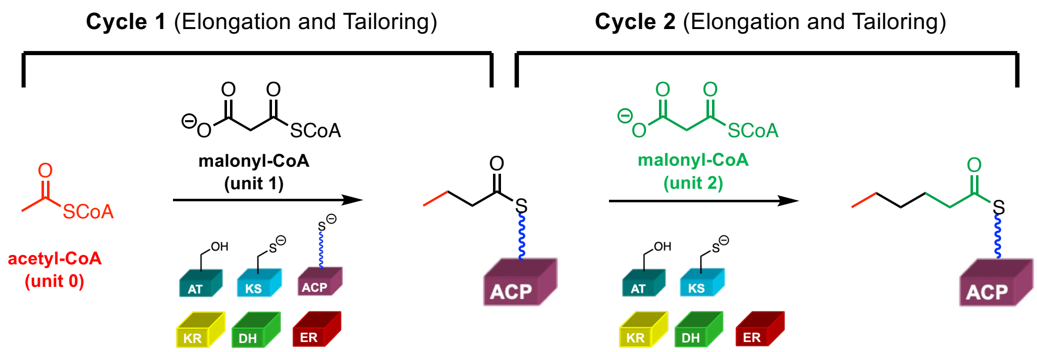


Ketoreductase

Dehydratase

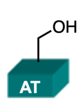
Enoyl Reductase

FAS Overview (2 Cycles)

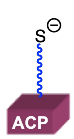


Fatty Acid Synthase Domains: Elongation

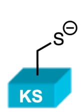
- Three protein domains are required to elongate the fatty acid by 1 unit (2 carbons):



AT Domain
Name: Acyl Transferase (**AT**)
Job: Select monomer unit and then transfer monomer to ACP
Active site: Serine



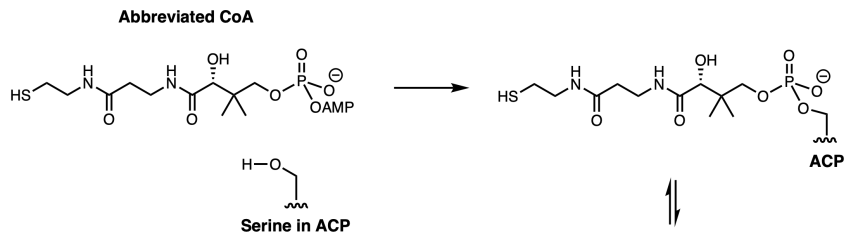
ACP Domain
Name: Acyl Carrier Protein (**ACP**)
Job: Carry growing chain between domains
Active site: Serine modified with long phosphopantetheinyl arm → thiol



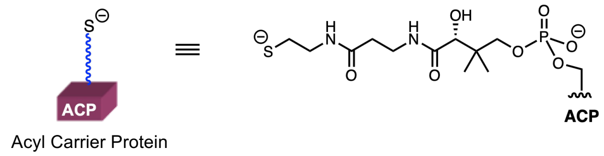
KS Domain
Name: Keto Synthase (**KS**)
Job: Accept growing chain from previous ACP and catalyze Claisen condensation with next monomer loaded ACP
Active site: Cysteine

Phosphopantetheinyl Arm

- The phosphopantetheinyl arm of the ACP domain comes from Coenzyme A:



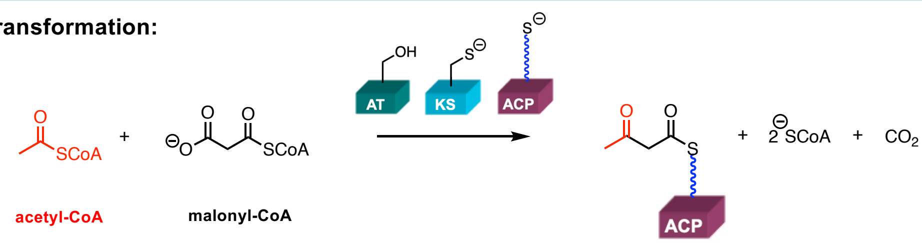
For biosynthesis, we will start with –SH deprotonated



Long phosphopantetheinyl arm in ACP carries the growing chain between domains

FAS Cycle 1: Elongation

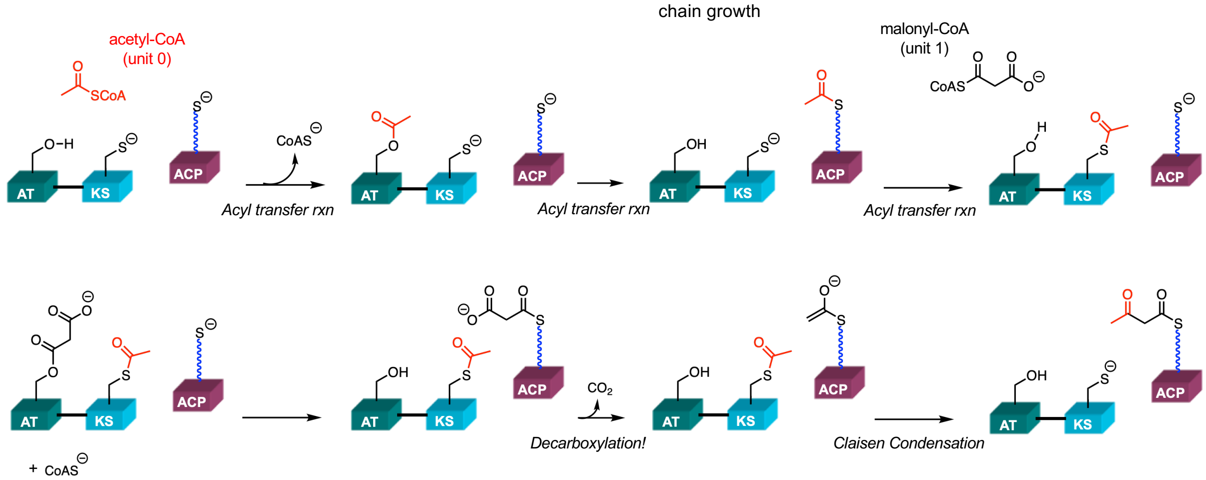
Overall transformation:



- Elongates the growing fatty acyl chain by **1 unit (2 carbons)**
- Uses **acetyl-CoA** as the primer (starter unit) and **malonyl-CoA** as the extender unit
- Forms the new C-C bond via a **decarboxylative Claisen condensations**, driven by CO_2 release

FAS Cycle 1: Elongation Steps

1. Load starter unit (acetyl-CoA) onto AT domain
2. ACP domain takes starter unit off of AT domain
3. KS domain accepts starter unit from previous ACP
4. AT Domain selects extender unit 1: malonyl-CoA
5. ACP domain takes extender unit off of AT domain
6. KS domain catalyzes decarboxylative Claisen Condensation: chain growth

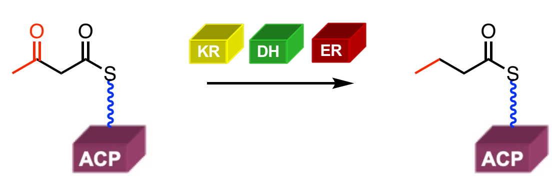


First elongation cycle completed!

FAS Cycle 1: Tailoring

- Tailoring steps reduce the ketone of the previous unit to a methylene group

Overall transformation:

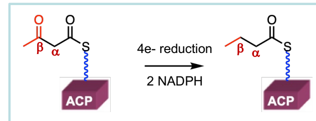


Poll Everywhere:

1. How many equivalences of NADPH are required?
2. Which carbonyl is reduced in FAS cycle 1?

Fatty Acid Synthase Domains: Tailoring

Overall:



- Three protein domains are required for tailoring steps:

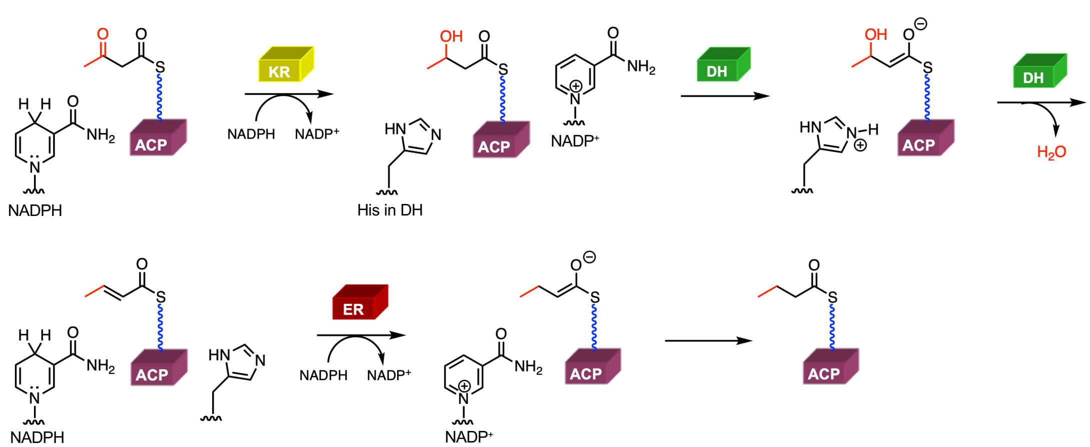
KR Domain
Name: β-Keto-Reductase (**KR**)
Job: Reduce the carbonyl of **the previous unit** to a hydroxyl
Active Site: Requires NADPH cofactor to provide hydride equivalent

DH Domain
Name: Dehydratase (**DH**)
Job: Form α,β–unsaturated double bond by removing hydroxyl as water
Active Site: Histidine as a general base

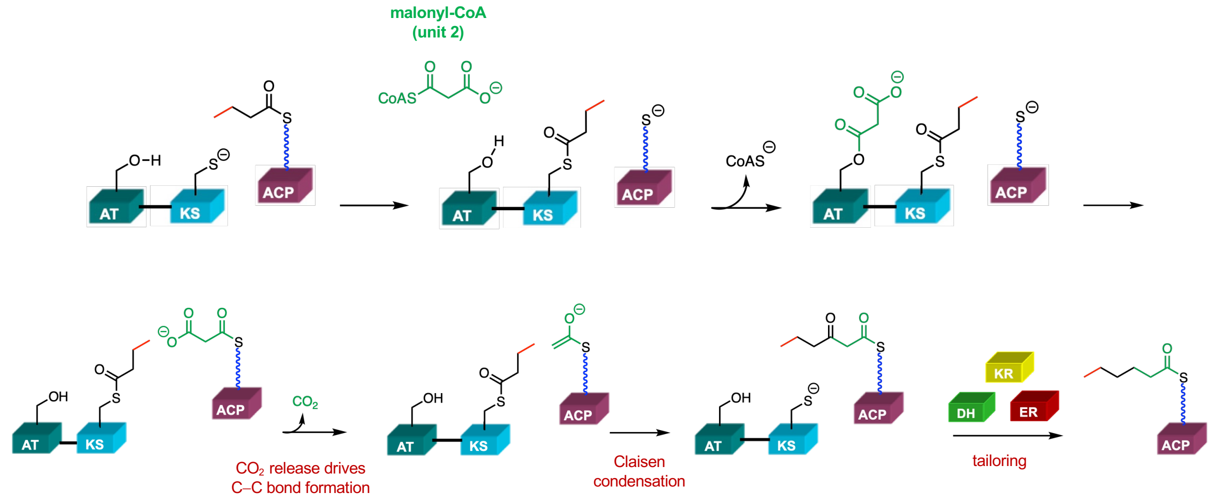
ER Domain
Name: Enoyl Reductase (**ER**)
Job: Reduce α,β–double bond to saturated single bond
Active Site: Requires NADPH cofactor to provide hydride equivalent

FAS Cycle 1: Tailoring Steps

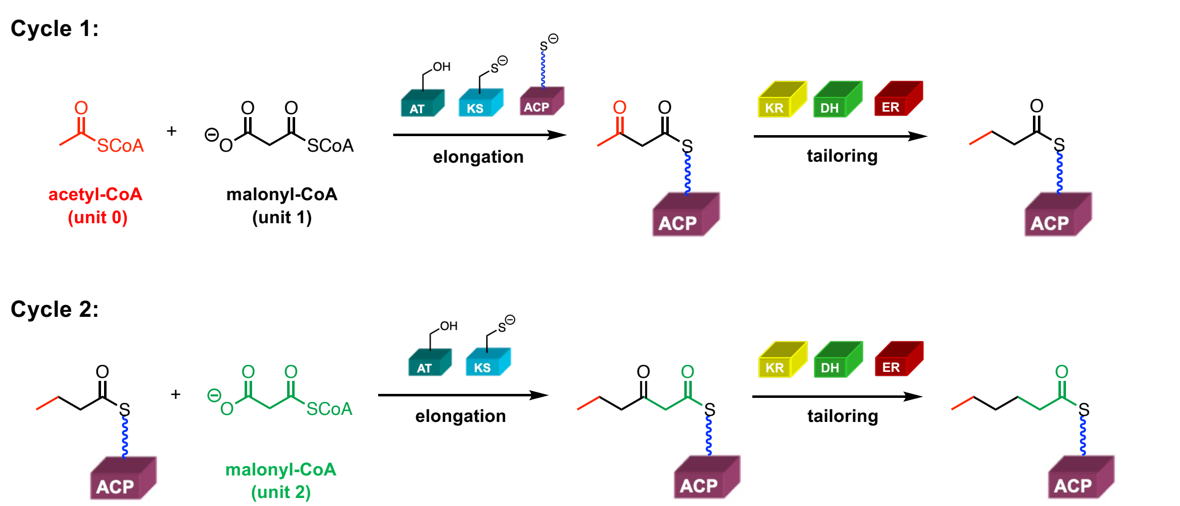
7. KR Domain reduces the preceding ketone of unit 0 (from acetyl-CoA)
8. DH Domain catalyzes elimination of water to form α,β–unsaturated thioester (**Poll Everywhere**: by which mechanism?)
9. ER domain reduces the α,β–unsaturated thioester to give a β–methylene!



FAS Cycle 2: Repeat Elongation and Tailoring



FAS Summary (2 Cycles)



Lecture 13 Objectives

- Learn the “business end” structures of acetyl-CoA and malonyl-CoA
- Draw the curved-arrow mechanism for a Claisen Condensation (“normal” and decarboxylative)
- Explain why nature generally employs thioesters, instead of oxoesters, as acyl transfer electrophiles and enolate precursors.
- Describe the role of the AT, ACP, and KS domains in a typical fatty acid synthase
- Identify the key amino acid residue or functional group that enables chemistry in the AT, ACP, and KS domains.
- Review an E1cb elimination reaction and the orbitals involved.
- Be able to write the mechanisms for all of the different FAS assembly line domains (AT, ACP, KS, KR, DH, ER, TE)
- By looking at a chemical structure, determine which domains (AT, ACP, and KS) would be required for each two carbon unit in the synthesis of a fatty acid